MILITARY SPECIFICATION

MASK, OXYGEN, PLASTIC, DISPOSABLE, ORONASAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of disposable oronasal mask of flexible plastic complete with tube and inlet connector for the purpose of administering oxygen.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center Drawing (DPSC)

No. 24176 - MASK, OXYGEN, PLASTIC, DISPOSABLE

(Copies of specifications, standards, drawing, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-412 Tension Testing of Vulcanized Rubber.

(Application for copies should be addressed to The American Society for Testing and Materials, 116 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The oxygen mask shall be made of high quality, pliable, translucent polyethylene or suitable transparent plastic. There shall be a rotatable turret type tubing connector made of rigid polyvinyl chloride plastic attached to the center of the mask and to which there is permanently attached a pliable, transparent length of medical grade polyvinyl chloride tubing. The tubing at its distal end shall have a polyvinyl chloride female connector permanently attached to it. The elastic head band shall be made of a suitable elastic material.

3.2 Style, design, and dimensions. The mask shall be designed to be usable on a wide variety of facial shapes and sizes and cover the oronasal area without extending under the chin. A plastic tubing connector shall be present on the mask at a location corresponding to the zone beneath the nose and near the mouth of the wearer. The turret-type tubing connector shall be rotatable 360° in either direction, and shall have a baffle diffuser for directing oxygen flow across the interior surface of the mask. An exhaust opening about one-half inch in diameter shall be present in the mask. The tubing shall be a minimum of 60 inches in length. The distal end shall have a tapered, female connector for attachment to the standard nipple of a regulator from an oxygen supply. An elastic head band shall be present and attached to the face mask. Free end of the headband shall be dipped in cement to prevent fraying. The entire assembly shall be in accordance with Defense Personnel Support Center Drawing No. 24176.

3.2.1 Tensile strength and elongation. There shall be a minimum tensile strength of 1900 pounds per square inch (psi) and ultimate elongation of 225 percent when the tubing is tested as specified in 4.4.1.

3.3 Safety (skin contact) test. The plastic mask material shall not cause skin irritation when tested as specified in 4.4.2, nor shall there be a delayed skin reaction 24 hours after removal of the patch.

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3.4 Certificate of safety (skin contact) test. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested as specified in 3.3 and found to be nondermatitic. This certificate shall be signed by a doctor of medicine and shall include the specific tests performed and the test results. For destination inspection, certificates shall be submitted by the contractor directly to Defense Personnel Support Center, Directorate of Medical Materiel, Technical Services Branch, Technical & Quality Assurance Division, 2800 South 20th Street, Philadelphia, PA 19101, ATTN: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government representative.

3.5 Security of mask-tube assembly. There shall be no evidence of separation at the junction of the tubing and the mask when subjected to stress as required in 4.4.3.

3.6 Workmanship. Workmanship shall be first class throughout. The oxygen mask, the tubing and other components shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the constractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the plastics may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the item identification, the National Stock Number, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 *Sampling*.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as specified in table I. The unit of product for examination purposes shall be one oxygen mask assembly.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

TABLE II. Sampling for tests.

Component	Characteristic	Reqmt.	Test procedure	Insp. level	AQL (percent defective)
Tubing	Tensile Strength	3.2.1	4.4.1	S-2	1/
Mask	Safety	3.3	4.4.2	2/	1/
Assembly	Security	3.5	4.4.3	S-2	1/

1/ Acceptance number shall be zero for all sample sizes.
2/ A minimum of five masks shall be tested from each lot.

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4.3 Examination. The oxygen mask assembly shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however shall not be restricted to defects listed in table III. Any oxygen mask assembly in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for test sample, the lot represented by the sample shall be rejected.

TABLE III. Classification of defects.

Category	Defect
Critical	None defined.
Major	
101	Mask and tube not assembled and ready to use.
102	Component part missing or damaged.
103	Plastic brittle or aged.
104	Tubing connector not rotatable at mask.
105	Plastic tubing not green tinted to identify oxygen use.
106	Mask surface at face contact is not smooth.
107	Foreign material molded in plastic.
108	Foreign matter on parts.
109	Tubing is tacky to touch.
Minor	
201	Tubing kinked.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Tensile strength and elongation. The tensile strength and ultimate elongation testing shall be performed in accordance with ASTM D-412.

4.4.2 Safety (skin contact) test. Apply a 0.5 inch by 0.5 inch piece of the face mask to the skin on the body side of the upper arm of five volunteers. The patch of plastic shall be held in place by means of a strip adhesive tape of a width not exceeding the width of the plastic sample. The plastic sample shall be allowed to remain in contact with the skin for 24 hours. The patch shall be removed from the contact area and no reaction shall be visible. Within an additonal 24 hours there shall be no delayed reaction on any of the subjects. If the patch unduly irritates the skin, the patch shall be removed immediately.

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4.4.3 Security of mask-tube assembly. A two-pound weight shall be attached to the tubing while the mask is held in a clamp for 10 minutes.

4.5 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing fifty masks, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Individual package. Each mask shall be thoroughly clean and packaged along with instructions for use in a snug fitting, clear, 3 mil, heat sealed polyethylene bag. The mask and tubing shall be packaged to prevent the tubing and mask from being kinked or deformed.

5.1.2.2 Unit package. Fifty individual packages shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, type CF, class domestic. Closure of the box shall conform to method I of PPP-B-636.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A. The unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, waterproofing and strapping shall be as specified in the appendix of the box specification.

5.2.2 Level B. Two unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure of the box shall conform to method I or II as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Individual package. Each package shall be marked to identify the contents and quantity. Marking may be printed on the bag or on an insert placed within the bag in such a manner so as to permit ready identification.

5.3.2 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The oxygen mask and attached tubing shall be used to administer oxygen therapy to a patient.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number, title and date of this specification.
(b) Applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

NSN	Item Identification
6515-00-888-6122	MASK, OXYGEN, Plastic, Disposable, Oronasal, 50s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:
Defense Supply Agency - DM

Project No. 6515-2024

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Philadelphia, PA 19101

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PHILADELPHIA, PA 19101

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-M-37553 - MASK, OXYGEN, PLASTIC, DISPOSABLE, ORONASAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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